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MILITARY SPECIFICATION

RADIO SET: AN/PRC-75B

This specification is approved for use by the U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of small, lightweight, portable, ultra-high frequency (UHF) radio set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILTIARY

MIL-T-704 - Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5820

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MIL-P-15024	-	Plate, Tags and Bands for Identification of Equipment.
MIL-E-16400	-	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for.
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts), Packaging and Packing of.
MIL-H-49078	-	Handset H-250( )/LL

STANDARDS

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MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-188	-	Military Communication System Technical Standards.
MIL-STD-449	-	Radio Frequency Spectrum Characteristics, Measurement of.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interferences Characteristics, Measurement of.
MIL-STD-470	-	Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-781	-	Reliability Test Exponential Distribution.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.

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DOD-STD-1686 - Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies, and Equipment.

HANDBOOKS

MILITARY

DOD-HDBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. MARINE CORPS

671-3412 - Production Test Requirements for Receiver- Transmitter Type AN/PRC-75B and RT-976B/PRC-75

(Copies of specification, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D3951-82 - Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), prior to submission of any complete production items for acceptance, the contractor shall furnish five first articles for determination of conformance to this specification. Examination and tests shall be those specified herein. Approval of the first articles by the acquiring activity shall not relieve the contractor of his obligation to supply radio sets conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to the approval of the contracting officer. The first articles shall not be included in the quantity to be delivered to the Government.

3.2 Parts, materials and processes. Parts, materials and processes shall be in accordance with MIL-E-16400 except as otherwise specified herein.

3.2.1 Tropicalization. Tropicalization of parts, materials and processes shall be in accordance with the requirements of MIL-STD-454. Hermetically sealed containers shall have at least 100 microinches of gold plating on kovar leads.

3.2.2 Fungus resistance. Wherever possible, only fungus nonnutrient material and parts shall be used. All other materials and parts used shall be treated as specified in MIL-STD-454 prior to use in the equipment. Moisture fungusproofing of the overall radio set shall not be permitted.

3.3 Design and construction. The radio set shall be designed and constructed as a portable tactical, crystal controlled, multichannel, amplitude modulated (AM), transmitting and receiving system. The radio set shall be operationally compatible with, but not restricted to operation with, AN/PRC-41, AN/PRC-41A, AN/ARC-51 and AN/ARC-55 radio sets on common frequencies. The radio set shall be compatible with certain existing secure voice equipments as specified herein. In the design and fabrication of the radio set, emphasis shall be

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placed on the application of solid state microelectronic techniques to achieve significant size and weight reductions and reliability improvements over conventional miniaturization techniques. Among the processes to be considered shall be silicon solid state circuits, pellet, micromodule and other circuit disciplines and thin film techniques involving various combinations of these processes into integrated active and passive networks. To the extent necessary to satisfy all of the specific requirements of this specification, the requirements of MIL-E-16400 shall apply. In order to assure compatibility with other communication systems and equipments employed by the military service, the radio set shall conform to MIL-STD-188. The radio set shall consist of one each of the following:

Item	Description
1.	Receiver-Transmitter (RT), RT976B/PRC-75
2.	Antenna, AS-2447/PRC-75
3.	Adaptor, Antenna, UG-2884/PRC-75
4.	Battery, dry, BA 3553/U
5.	Battery, Box CY6792/PRC-75
6.	Cable Assembly, TL-1371
7.	Cable Assembly, TL-1372
8.	Headset, H251/U
9.	Harness, Radio Set, ST-162/PRC-75
10.	Lanyard
11.	Case, Radio Set, CY-6729/PRC-75
12.	Handset, H-250/U

The RT shall be designed to operate from direct current (DC) supplies within the voltage limits (steady state and transient) specified herein.

3.3.1 Modular concept. Maintainability shall be used on a modular concept. The design shall be such that the following plug-in functional modules and chassis shall comprise the RT unit:

Main chassis and control head (1A1)
Voltage regulator (1A2)
Frequency synthesizer (1A3)
Intermediate frequency (IF)/Audio (1A4)
Radio frequency (RF) amplifier/modulator (1A5)
Secure voice relay (1A6)

3.3.2 Mechanical stability. All controls and adjustments shall be designed so that their position of adjustment will not be altered when the RT is subjected to the service conditions specified herein.

3.3.3 Solid state components. All active circuit elements shall be solid state devices. No vacuum tubes shall be used.

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3.3.4 Removable components. The design of the removable parts and assemblies shall be such that they may be removed from the RT either singly or in any combination, without unsoldering any connections. After being so phyisically removed from the RT, it shall be possible, with plug-in extension cables, to connect them with the main chassis to form a complete working unit for testing purposes. This shall be accomplished without necessitating the removal of any shafts, controls or other subassemblies other than the desired subassemblies.

3.3.5 Measurement of RF spectrum and plans for collection. Measurement of RF spectrum and plans for collection shall be in accordance with MIL-STD-449. Spectrum signature testing involving flight testing shall not be conducted by the contractor.

3.3.6 Controls locations. Except for the press-to-talk (PTT) button, all controls for the RT shall be on the top panel. The PTT button shall be on the front panel of the radio set.

3.3.7 Channels. The radio set shall be capable of being tuned to any one for 3,500 channels in the 225.00-megahertz (MHz) to the 399.95-MHz frequency spectrum. Channel frequency increments shall be 50 kilohertz (KHz). Selection of any one of the 3,500 channels shall be possible, utilizing no more than three frequency selection knobs on top of the radio set and without adding or changing crystals.

3.3.8 Frequency selection controls. Three frequency selection controls shall be provided to select any frequency in the 225.00- to 399.95-MHz range in 50-kHz increments. A stop shall be provided on each frequency selection control and each position on the controls shall be indented. The indentations shall be sufficiently pronounced to enable frequency selection in total darkness by counting indents from the stop. Panel illumination shall be provided and shall be available to the operator by utilizing a momentary contact switch.

3.3.9 Frequency readout. A digital readout indicating the operating frequency in MHz shall be provided as part of the RT.

3.3.10 Resetability. Resetability shall be such that upon rechanneling, the frequency shall be within 1 part in 10^5 of the original frequency.

3.3.11 Channel selection time. The time required to change any operating channel to any other operating channel by an operator wearing combat mittens shall not exceed 15 seconds.

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3.3.12 Channel "lock-on" time. The time required for the RT to become operational after any operating channel has been changed to any other operating channel shall not exceed 1 second.

3.3.13 Silent operation. The production of audible noise by operation of the RT controls shall be held to the minimum extent practicable. The received audio signal shall be excluded.

3.3.14 Warm-up time. Over the temperature ranges specified below, the RT shall be fully operational when given the following warm-up times after turn on:

Temperature Range	Warm-up Time
+10°C to +52°C	1 second
-30°C to +10°C	15 seconds

3.3.15 Continuous transmit operation. The transmitter shall be designed to be capable of 100 percent continuous duty at maximum RF power output for a period of 30 minutes. Protective devices shall be provided, where necessary, to prevent self-damage to the RT during this type of operation.

3.3.16 Case temperature. In an ambient temperature of 70°F (21.5°C) and normal pressure and with continuous transmit operation, the case temperature of the RT shall not exceed 100°F (37.8°C).

3.3.17 Microphonics and other noise. The RT shall be designed so that no microphonics or other noise shall exist in the radio set which would modulate the carrier to interfere with the received audio output.

3.3.18 Voltage variation. The RT shall be capable of operating, within the requirements specified herein, without damage to components over a supply voltage range of 22 to 30 direct current volts (VDC).

3.3.19 Combined low temperature and low voltage operation. The RT may operate with reduced performance when the input voltage is reduced to 18 volts (V), and ambient temperature is reduced to -30°C as specified herein.

3.3.20 RF source. The RF source shall be crystal controlled. All channel frequencies shall be derived from a single crystal. The source shall be common to both the transmitter and the receiver. If double conversion is to be used in the receiver, one additional crystal may be used. However, it shall not be necessary to change any crystals when changing the operating frequency.

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3.3.21 Audio gain control. An audio gain control shall be provided to vary the audio level to the external handset/headset. The audio output into a 500-ohm resistive load shall be continuously variable over a range of at least 50 decibels (dB).

3.3.22 Function selector switch. A function selector switch, having the following positions, shall be provided on the front panel:

(a) OFF - This position shall disconnect battery from the RT.

(b) T/R - This position shall provide transmit-receive operation of the RT.

(c) Direction finder (DF) - This position shall turn on the transmitter in a tone modulated mode.

3.3.23 Input power. The power input requirements shall not exceed the following at room ambient conditions:

Input voltage	+24 VDC (nominal)
Receive power	3.6 watts (W) at 24 VDC
Transmit power	15.6 W at 24 VDC

During secure voice operation, provision shall be made for the supplying of up to 2 W of additional DC input power to external equipment.

3.3.24 Fusing. One fuse shall be provided with an appropriate fuse holder and shall be placed in the circuit to the external battery. Fuse replacement shall be possible without removing the RT case cover.

3.3.25 Main connector facilities. The following facilities shall be provided on separate pins of the main connector:

External PTT

+24-V battery input

+24-V secure supply

Battery ground

Ground secure

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+6-V secure input (within the requirements of MIL-STD-188)

External microphone high and 1.2-V peak to peak secure voice input

Wideband audio output

Headset audio output (external headset/handset)

Provision shall be made to connect an H-250/U handset conforming to MIL-H-49078 in lieu of the external headset.

3.3.26 Basic conditions and definitions. For purposes of this specification, the following basic conditions shall apply, as pertinent to the specified performance requirements of the radio set, and shall be regarded as being in addition to those specified in MIL-STD-449. The conditions and definitions are specifically applicable to the measurement of sensitivity, spurious responses, selectivity and audio outputs.

3.3.26.1 Standard dummy antenna. The standard dummy antenna shall be a device, the electrical constants of which are such as to simulate the impedance characteristics of matched antennas. The dummy antenna shall be 50-ohm nonreactive resistance. The resistance added externally to the standard signal generator to simulate the 50-ohm resistance shall produce a total of 50 ohms when added to the effective resistance of the standard signal generator. The intent is that the standard signal generator produce the equivalent of the signal voltage induced by a radio field in an antenna which has the constants which are simulated by the dummy antenna. Hence, the microvolt reading obtained from the standard signal generator shall be the open circuit voltage in series with the dummy antenna.

3.3.26.2 Standard output signal-plus-noise to noise ratio. Standard output signal-plus-noise to noise ratio shall be 10 dB. The output signal-plus-noise to noise ratio shall be the ratio between a standard audio power output and the output due to noise with the modulation removed.

3.3.26.2.1 Standard audio output. Standard audio output shall be nominally 8 milliwatts (mW) of power dissipated in a 600-ohm nonreactive resistance connected across the external speaker terminals.

3.3.26.2.2 Standard gain. Standard gain shall be that gain which produces standard output single-plus-noise to noise ratio (see 3.3.26.2) with standard audio output.

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3.3.26.3 Standard modulation frequency. Standard modulation frequency shall be nominal 1,000 ±200 hertz (Hz).

3.3.26.4 Sensitivity. Sensitivity of the receiver is that characteristic which determines the minimum strength of a signal input for producing the desired ratio of output signal-plus-noise to noise specified in 3.3.26.2 at the prescribed level of 3.3.26.4.1.

3.3.26.4.1 Standard sensitivity. Standard sensitivity shall be the number of microvolts of pure RF energy (applied at the antenna input through a standard dummy antenna) required to produced not less than 8 mW at the audio output terminal with standard signal-plus-noise to noise ratio.

3.3.26.5 Selectivity. Selectivity of the RT is that characteristic which determines the extent to which it is capable of differentiating between the desired resonant frequency signal and RF signals at other frequencies.

3.3.26.5.1 Weak signal selectivity. Weak signal selectivity shall be the degree to which the radio set is capable of differentiating between the desired signal and interfering signals of frequencies relatively close to the desired signal frequency. It shall be expressed in terms of the attenuation in dB, which the RT introduces to unmodulated RF removed from resonance, with respect to a 4-microvolt input signal at the desired frequency.

3.3.26.6 Spurious responses.

3.3.26.6.1 Image rejection. Image rejection shall be the degree to which the RT is capable of rejecting signals which, in combination with the fundamental or any harmonic of the conversion oscillator(s), produce frequencies which are amplified by the IF amplifiers and result in spurious responses. It shall be expressed in terms of the attenuation, in dB, which the RT introduces to the image frequencies with respect to a 4-microvolt input signal at the desired channel frequency.

3.3.26.6.2 IF rejection. IF rejection shall be the degree to which the RT is capable of rejecting signals having the frequencies of the IF amplifiers. Intermediate rejection shall be expressed in terms of the attenuation, in dB, which the RT shall introduce to the IF with respect to a 4-microvolt signal at the desired channel frequency.

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3.3.26.7 Maximum audio distortion. Maximum audio distortion shall not exceed 15 percent, as measured by a distortion analyzer, when input signals are between 3 microvolts and 2000 microvolts, modulated at 30 percent, at 1000 Hz modulation frequency and an output power of at least +9 dB mW.

3.3.27 Transmitter.

3.3.27.1 Frequency range. The transmitter shall be capable of operating on any one of 3,500 channels in the frequency range from 225.00 to 399.95 MHz.

3.3.27.2 RF power output. Under standard conditions specified herein and using the battery supplied as part of the radio set, the tranmitter shall be capable of delivering not less than 1 W RF average power into a 50-ohm nonreactive load. The RF carrier shall be modulated 90 percent at 1,000 Hz for this requirement. At temperatures below 0°C but with normal supply voltage of 24 VDC, the transmitter RF power output and percentage modulation capability shall not degrade by more than 3 dB below the limits specified for standard conditions. Under conditions of low supply voltage (18 V) only, the transmitter RF power output and percentage modulation capability shall not degrade by more than 3 dB below the limits specified for standard conditions. Under conditions of low supply voltage (18 V) and at temperatures below 0°C, the transmitter RF power output and percentage modulation capability shall not degrade by more than 6 dB below the limits specified for standard conditions.

3.3.27.3 Frequency accuracy. Under any natural combination of service conditions specified herein, the frequency accuracy shall be such that the deviation from the nominal frequency shall not exceed ±4 kHz.

3.3.27.4 Frequency stability. The frequency stability of the transmitter shall be equal to or better than 1 part in 10^5 per day under any natural combination of service conditions.

3.3.27.5 Harmonic and spurious radiated and conducted interference. When use is made of the flexible blade antenna, the radiated interference levels of harmonics shall be at least 30 dB below the unmodulated carrier level at the operating frequency. All other spurious radiation shall be at least 60 dB below the carrier level at the operating frequency. When use is made of the antenna adaptor provided as part of the radio set, the antenna conducted interference levels of all harmonics of the operating frequency shall be at least 40 dB below the unmodulated carrier level at the operating frequency.

3.3.27.6 Modulation. The transmitter modulation shall be as specified herein.

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3.3.27.6.1 Modulation capability. The transmitter shall be capable of the following modes of modulation.

(a) Normal voice mode. The modulator section of the transmitter shall be capable of modulating the carrier at least 30 percent with a standard modulation frequency audio input at a level of 65 dB above 10^{-16} W per square centimeter at standard temperature and pressure provided to the audio transducer, under all service conditions. Not less than 90 percent modulation shall be maintained over a range of audio input levels from 75 dB above 10^{-16} W per square centimeter to 100 dB above 10^{-16} W per square centimeter. When driven from an external source (dynamic-type microphone), the modulator section of the transmitter shall be capable of modulating of the carrier at least 30 percent with a standard input signal whose level is a minimum of minus 56 dBm and 1000 Hz input. Input impedance of the modulator section shall have a nominal value of 125 ohms. A current source of 55 ±20 percent milliamperes shall be available for operation of an external secure voice device (strapped option).

(b) Secure voice mode. The modulator section of the transmitter shall be capable of 90 percent average modulation as viewed on an oscilloscope with a secure voice input signal of 1.2 V peak to peak at a frequency of 18.75 kHz. Input impedance of the modulator section shall have a nominal value of 125 ohms. During secure voice operation, the internal microphone shall be disabled. A minimum attenuation of 120 dB to analog signals from the internal microphone shall be provided.

(c) Secure voice ±6 V input. The modulator section of the transmitter shall be capable of 90 percent average modulation as viewed on an oscilloscope with a secure voice standard input signal of ±6 V at a frequency of 18.75 kHz. A separate modulator input of at least 1,000 ohm input impedance shall be provided.

3.3.27.6.2 Secure voice mode system frequency response. The overall system frequency response shall be as shown by Figure 1. This response shall be measured utilizing two AN/PRC-75 Radio Sets operating back to back in the secure voice mode. A 12-V peak to peak sinusoidal signal shall be inserted into the modulator of, and transmitted by, RT No. 1. RT No. 2 shall be sampled directly at the wideband audio output. For the purpose of this measurement, sufficient attenuation shall be provided between RT's to attain a receiver input RF signal of between 10 and 100 microvolts. The above test shall be repeated using a 1.2 V peak to peak sinusoidal signal connected into the microphone high and 1.2 V peak to peak signal into the secure voice terminal of the main connector.

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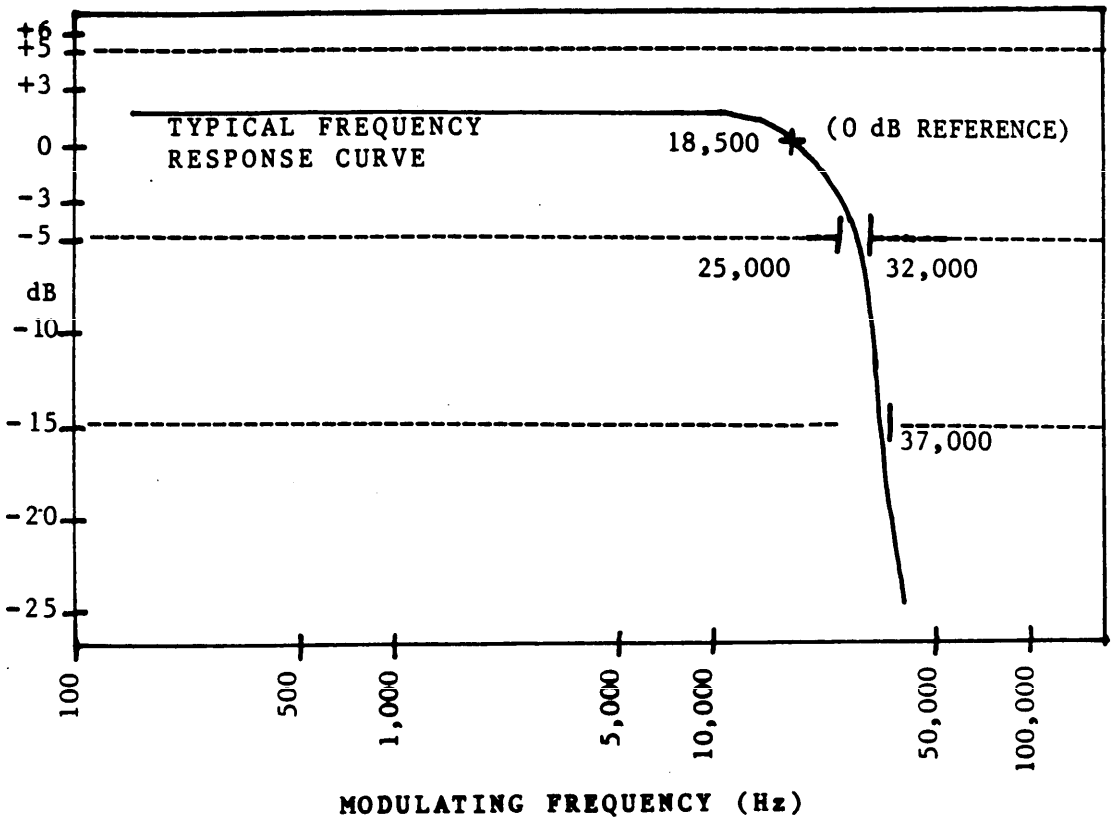


FIGURE 1. Overall system frequency response limits.

3.3.27.6.3 Modulation frequency response. The transmitter frequency response for the following modulation mode shall be as specified herein:

(a) Normal voice mode. The audio frequency response of the modulated envelope shall not vary more than +2, -6 dB relative to the response at 1,000 Hz from 300 Hz to 2,700 Hz using a dummy microphone having an impedance of 500 ohms resistive in place of the internal microphone.

(b) Secure voice mode (+/-6 V or 1.2-V input). The frequency response curve of the modulated envelope shall not cross the limits shown in Figure 2. The internal or external microphone shall be excluded from this requirement.

3.3.27.6.4 Envelope distortion. Under standard conditions specified herein, the total envelope distortion shall be less than 10 percent in the frequency range from 300 Hz to 2,800 Hz with the RF carrier modulated 64 percent. The internal or external microphone shall be excluded from this requirement and shall be replaced by resistive dummy microphones of equivalent nominal impedance and output voltage.

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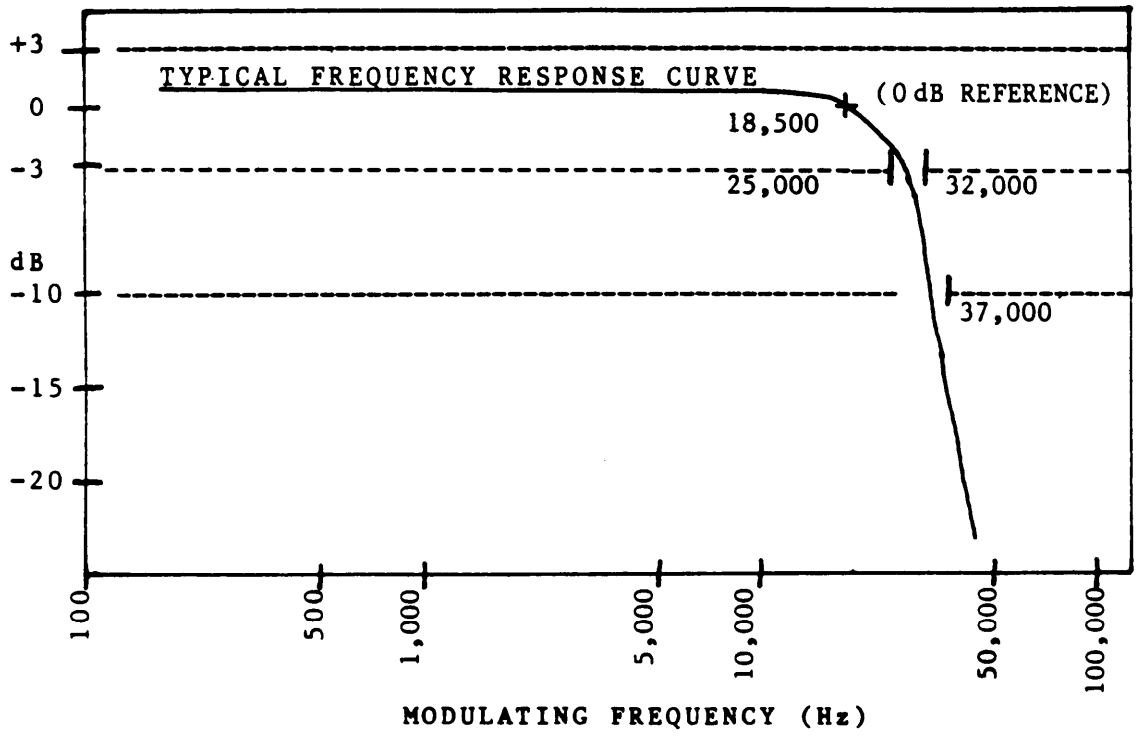


FIGURE 2. Modulator or demodulator frequency response limits.

3.3.27.6.5 Noise level. Under all service conditions specified herein, the modulation noise level shall be at least 30 dB below 90 percent modulation. Under service conditions of vibration testing, in accordance with Section 4 at vibration frequencies of isolated unavoidable mechanical resonances, the noise modulation of the transmitter carrier shall be at least 20 dB below the reference level. Under completion of vibration testing, the radio set shall meet the standard condition requirements of this specification. The internal and external microphones shall be excluded from this requirement and shall be replaced by resistive dummy microphones of equivalent impedance and output voltage.

3.3.27.6.6 Tone modulation (modulated continuous wave). The RT shall be capable of producing a tone modulated carrier for DF purposes. The frequency of the internally generated tone shall be the standard modulation frequency. In this mode of operation, the radio set shall provide at least 90 percent average modulation over the entire RF range covered by the RT.

3.3.27.7 Speech processing. Speech processing techniques such as preemphasis, deemphasis, compression and limiting shall be employed, at the discretion of the contractor, to provide maximum power output consistent with intelligibility and compatibility considerations and other requirements specified herein.

3.3.28 Receiver. The receiver shall be of the superheterodyne type.

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3.3.28.1 Frequency range. The receiver shall be capable of operating on 3,500 channels over the frequency range from 225.00 to 399.95 MHz.

3.3.28.2 Dynamic range. The receiver shall have a dynamic range of at least 86 dB, consistent with the requirements of 3.3.28.7.

3.3.28.3 Sensitivity requirement. The receiver sensitivity shall be as specified herein for the following modes:

(a) Normal voice mode. Under standard conditions specified herein, the sensitivity of the receiver shall be such that when operating the radio set below 320 MHz and an RF input of 4.0 microvolts modulated 30 percent at 1,000 Hz is applied through a standard dummy antenna, the audio output shall be at least 8 mW into a 600-ohm resistive load with a signal-plus noise ratio of at least 10 dB. When operating the radio set above 320 MHz, the sensitivity of the radio set shall be such that when a RF input of 5.2 microvolts modulated 30 percent at 1,000 Hz is applied, the audio output shall be at least 8 mW into a 600-ohm resistive load with a signal-plus-noise to noise ratio of at least 10 dB. The above requirement is equivalent to a closed circuit signal of 2 microvolts and 2.6 microvolts respectively. At temperatures below 0°C but with normal supply voltage, the receiver sensitivity and audio output shall not degrade more than 3 dB below the standard conditions requirement. Under conditions of low supply voltage (18 V) only, the receiver sensitivity and audio output shall not degrade more than 3 dB below the standard conditions requirement. Under conditions of low supply voltage (18 V) and at temperatures below 0°C, the receiver sensitivity and audio output shall not degrade more than 6 dB below the standard conditions requirements.

(b) Secure voice mode. Under standard conditions specified herein, the sensitivity of the receiver shall be such that when an RF input of 10 microvolts (closed circuit) modulated 30 percent at 1,000 Hz is applied, the audio output at the secure voice terminals shall be at least 0.25 V root mean square (RMS) across a 22,000-ohm load with a signal-plus-noise to noise ratio of at least 10 dB. At temperatures below 0°C, but with normal supply voltage, the receiver sensitivity shall not degrade more than 3 dB below the standard conditions requirement. Under conditions of low supply voltage (18 V) only, the receiver sensitivity shall not degrade more than 3 dB below standard conditions requirement. Under conditions of low supply voltage (18 V) and at temperatures below 0°C, the receiver sensitivity shall not degrade more than 6 dB below the standard conditions requirement.

3.3.28.4 Receiver IF selectivity. The IF selectivity of the receiver shall fall within the limits in Table I.

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TABLE I. Intermedidate frequency bandwidth

LEVEL AS COMPARED WITH STANDARD MODULATING SIGNAL	INTERMEDIATE FREQUENCY BANDWIDTH
-3.0 dB	NO LESS THAN 60 kHz
-20 dB	NO LESS THAN 74 kHz
-30 dB	NO LESS THAN 80 kHz
-60 dB	NO MORE THAN 120 kHz

3.3.28.5 Image rejection. At operating frequencies below 320 MHz, the receiver image rejection shall be at least 60 dB relative to a 4-microvolt RF input to all image frequencies. At operating frequencies above 320 MHz, the receiver image rejection shall be at least 50 dB relative to a 4-microvolt RF input to all image frequencies.

3.3.28.6 IF rejection. Rejection of signals at the IF shall be at least 80 dB relative to a 4-microvolt RF input at any operating frequency.

3.3.28.7 Automatic gain control (AGC) characteristics. The AGC action during normal voice mode shall be such that the audio output power shall vary no more than 3 dB with an input signal variation of from 7 microvolts to 100,000 microvolts. The input signal shall be modulated 30 percent at 1,000 Hz.

3.3.28.8 Audio distortion. Under standard conditions specified herein, in either normal or secure voice mode, with an input signal of 1,000 microvolts modulated 30 percent at 1,000 hz, the audio distortion shall not exceed 15 percent when delivering specified audio input into specified resistive loads:

3.3.28.9 Receiver audio frequency response. The audio frequency response for the receiver shall be as follows:

(a) Normal voice mode. The audio response for the receiver with a 1,000-microvolt signal, modulated 30 percent by modulation frequencies from 300 to 2,700 Hz, shall not vary more than +2 or -6 dB relative to the response at standard modulation frequency.

(b) Secure voice. The audio response for the receiver with 1,000-microvolt signal modulated 30 percent shall be in accordance with the requirements of Figure 2.

3.3.28.10 Error rate. When operating in the secure voice mode, the error rate shall not exceed 5 bits per thousand.

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3.3.29 Antenna. A flexible removable antenna, mounted on the RT, shall be provided. The antenna design shall be such that the length need not be adjusted for any operating frequency within the 225.00 to 399.95 MHz range. The flexible antenna shall have sufficient rigidity to support its own weight, but shall be flexible enough to pass under rigid obstacles located 12 inches or more above the RT. The antenna shall be storable within the confines of the space allotted to the RT.

3.3.29.1 Antenna system efficiency. The efficiency of the antenna shall be such that at any frequency of operation within the 225.00 to 399.95 MHz range, the radiation efficiency shall be at least 50 percent of the efficiency of a tuned quarter wave antenna without any physical adjustment to the length of the antenna.

3.3.30 Antenna adaptor. An antenna adaptor which will attach to the side of the RT in place of the flexible blade antenna shall be provided as part of the radio set. The antenna adaptor shall permit the radio to operate with external antennas such as antenna (flexible), AS-2117/PRC-66, or antenna (dipole), AS-1404/PRC-41, or to operate with higher power amplifier units. The antenna adaptor shall contain a low-pass filter to provide additional attenuation to harmonic and spurious antenna conducted interference as detailed in 3.3.27.5.

3.3.31 External headset. A suitable lightweight external headset such as H-251 ( )/U shall be provided. The headset will not be required to pass the environmental requirements of this specification; however, the headset shall be subjected to the same test conditions as the rest of the radio set and the resulting data shall be supplied to the acquiring activity.

3.3.32 Battery. The radio set shall operate from a self-contained expendable battery (pack) power supply. The battery shall consist of alkaline, or equivalent, nonrechargeable cells. Nominal battery pack voltage shall be 24 V.

3.3.32.1 Battery life. Under standard conditions specified herein, a fresh battery shall be capable of providing the total input power requirements for a period of at least 8 hours of normal voice mode of operation when the radio set is operated at a duty cycle of 9 to 1 (9 minutes receive and 1 minute transmit). This requirement shall be restricted to the use of the internal microphone only.

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3.3.32.2 Reverse polarity protection. The radio set design shall include a protective device to insure that reversal of battery polarity shall cause no damage to the battery or RT. Additionally, the battery supply shall have terminations to prevent incorrect battery connection by the operator.

3.3.32.3 Battery case. The battery shall be housed in a reusable case which shall be easily detachable from the RT without exposing electronic parts to possible damage.

3.3.33 Connecting cables. Any required interconnections between system components shall be by means of ruggedized flexible cables with strain relief and, where practicable, with connectors which will disconnect before cable failure in the event the cable is subjected to unexpected strain. All cables shall be in accordance with requirement 10A of MIL-STD-454.

3.3.34 Connectors. All external, internal and RF connectors shall, as far as practicable, be in accordance with requirement 10A of MIL-STD-454.

3.3.35 Carrying harness. The carrying harness shall provide support for the RT and battery pack under the operator's arms. The harness shall also provide support for any required connecting cables. The RT and battery pack shall be easily removable from the carrying harness while the operator is wearing the harness.

3.3.36 Lanyard. A lanyard shall be provided as part of the radio set. The lanyard shall allow the RT to be suspended from the operator's neck, permitting full use of both hands. Two lanyard attachment points shall be provide on opposite sides of the top of the RT case.

3.3.37 Transit case. A waterproof transit case shall be provided to enclose and protect the RT and all accessories during any form of field transportation or storage. It shall be capable of protecting all the enclosed components under the conditions indicated in 3.6.3 and 3.6.4. The transit case shall be of rugged construction, lightweight material, compartmentalized and contain shock absorbing material to provide maximum protection to the components enclosed. The transit case shall be equipped with an easily removable cover and a cover retaining bracket. Folding handles shall provide hand holds for carrying.

3.3.38 Interchangeability. The components, parts and accessories of the radio set shall conform to the interchangeability requirements of requirement 7 of MIL-STD-454. The modules shall also be checked by substitution in the Government-loaned AN/PRC-75A and AN/PRC-75B (see 3.12).

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3.4 Reliability program.

3.4.1 Reliability requirements. The radio set shall possess an inherent mean time between failure (MTBF) of not less than 2,000 hours base on uncontrolled ground environment conditions in a man-pack configuration as demonstrated by 4.6.5. A failure shall be defined by MIL-STD-781.

3.4.2 Longevity. The radio set shall have a longevity or minimum operating life of 10,000 hours with reasonable servicing and replacement of parts before wear-out failures occur or the MTBF become less than the value specified herein. Parts requiring replacement within this interval and the predicted life of such parts shall be reported by the contractor to the acquiring activity, as specified by the contract or order.

3.5 Maintainability program.

3.5.1 Mean time to repair (MTTR). The RT shall possess a MTTR at the module replacement level (field level) of not greater than 30 minutes.

3.6 Performance. The radio set shall be capable of the following performance requirements.

3.6.1 Electrical stability. The radio set, except for the battery power supply, shall operate as specified herein for at least 2,000 hours, either continuously or intermittently, independent of component failure, without necessity for readjustment of any controls other than those normally accessible to the operator. Normal realignment in the process of equipment repair is excepted.

3.6.2 Standard conditions. The following shall be used as a basis for establishing normal performance requirements and for conducting tests of the radio sets:

(a)	Barometric Pressure	28 to 32 inches Mercury
(b)	Temperature	15°C to 35°C
(c)	Humidity	50 percent ± 30 percent
(d)	Supply Voltage	24 ± 0.5 VDC

3.6.3 Service conditions. The radio set shall be capable of operating under the following service conditions:

(a)	Ambient Temperature Range	-30°C to +52°C

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(b)	Ambient Humidity Range	0 to 100 percent Relative
(c)	Low Temperature	Method 502, Procedure 1, MIL-STD-810
(d)	High Temperature	Method 501, Procedure 1, MIL-STD-810
(e)	Altitude	Method 504, Procedure 1, MIL-STD-810
(f)	Temperature Shock	Method 503, Procedure 1, MIL-STD-810
(g)	Leakage	Method 512, Procedure 1 MIL-STD-810
(h)	Dust	Method 510, Procedure 1, MIL-STD-810
(i)	Shock	Method 516, Procedure II & V, MIL-STD-810, Drop 42 inches to flat surface of solid wood (1 inch fur) backed by at least 3 inches of concrete along each of three perpendicular planes
(j)	Vibration	Method 514-2 VIII, Procedure Curve Y of Figure 514-2-6 and Part 2 of Procedure XI of MIL-STD-810
(k)	Rain	Method 506, Procedure 1, MIL-STD-810
(l)	Humidity	Method 507, Procedure 1, MIL-STD-810
(m)	Fungus	Method 508, Procedure 1, MIL-STD-810
(n)	Salt Fog	Method 509, Procedure 1, MIL-STD-810

3.6.4 Storage conditions. The radio set shall be capable of being stored under the following conditions without degradation.

(a) Ambient Temperature Range -54°C to +68°C

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(b) Altitude	Sea Level to 50,000 ft.

3.7 Electromagnetic interference (EMI) suppression. The supplier shall establish an EMI control plan for the radio set in accordance with requirements of MIL-STD-461.

3.8 Dimensions. The radio set components shall not exceed the following dimensions (excluding any knobs and protrusions):

Receiver-Transmitter	1.625 by 4.175 by 8.063 inches
Battery (with case)	2.3 by 4.7 by 6.6 inches
Antenna Adaptor	1.375 by 1.75 by 3 inches

3.9 Weight. The radio set components shall not exceed the following weights:

Receiver-Transmitter	3.9 pounds maximum
Battery	4.5 pounds maximum
Antenna Adaptor	0.25 pounds maximum
Harness, Radio Set	1.5 pounds maximum

3.10 Finish. The radio set shall be finished in accordance with the requirements of MIL-T-704, type F or G as applicable. Exterior color shall be green 383.

3.11 Identification marking. Size permitting, modules and module subassemblies shall be identified with the following information rubber stamped or silk screened on a fixed surface, or on a protected tag affixed in a manner that does not impair operation of the assembly and will not easily be torn off:

(a) Nomenclature
(b) Name, if applicable
(c) Additional information such as Federal Stock Number.

3.11.1 Size limited marking and identification. Where limited size of modules and module subassemblies precludes conformance with 3.11, the contractor shall submit to the contracting officer a method of marking which will ensure that such items may be identified.

3.11.2 Nameplates and instruction plates. Nameplates and instruction plates shall conform to the requirements of MIL-P-15024.

3.12 Government-loaned property. The Government shall loan an AN/PRC-75A and an AN/PRC-75B radio set for use by the contractor to determine design requirements and for test purposes as specified herein.

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3.13 Workmanship. The radio set covered by this specification, including all parts and accessories, shall be manufactured in accordance with the requirements of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchases order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3)
(b) Production (see 4.4)
(c) Sample (see 4.5)

4.3 First article inspection. The first article shall be subjected to the tests indicated for first article in Table II. The details of the test to be performed shall be called out in the first article test plan.

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TABLE II. Tests.

TEST	REQ. PARA	TEST PARA	FIRST ART. TESTS	PRODUC-TION TESTS	SAMPLE TESTS	TEST NO.
MECHANICAL	3.2 and 3.3.2	4.6.1	X	X	X	1
INPUT POWER	3.3.23	4.6.2	X	X	X	2
RADIO FREQ. POWER OUTPUT	3.3.27.2	4.6.2	X	X	X	3
MODULATION CAPABILITY	3.3.27.6.1	4.6.2	X	X	X	4
FREQUENCY ACCURACY	3.3.27.3	4.6.2	X	X	X	5
AUDIO DISTORTION	3.3.28.8	4.6.2	X	X	X	6
DYNAMIC RANGE	3.3.28.2 3.3.28.7	4.6.2	X	X	X	7
AUDIO FREQ. RESPONSE	3.3.28.9	4.6.2	X	X	X	8
AUDIO GAIN CONTROL	3.3.21	4.6.2	X	X	X	9
RADIO FREQ. SENSITIVITY	3.3.28.3	4.6.2	X	X	X	10
INTERMEDIATE FREQUENCY SELECTIVITY	3.3.28.4	4.6.2	X	X	X	11
IMAGE REJECTION	3.3.28.5	4.6.2	X		X	12
INTERMEDIATE FREQUENCY REJECTION	3.3.28.6	4.6.2	X		X	13
CARRIER NOISE LEVEL	3.3.27.6.5	4.6.2	X		X	14
SYSTEM FREQUENCY RESPONSE	3.3.27.6.2	4.6.2	X		X	15
MODULATION FREQUENCY RESPONSE	3.3.27.6.3	4.6.2	X	X	X	16
TONE MODULATION (MODULATED CON-TINUOUS WAVE)	3.3.27.6.6	4.6.2	X	X	X	17
ENVELOPE DISTORTION	3.3.27.6.4	4.6.2	X	X	X	18
CASE TEMPERA-TURE	3.3.16	4.6.3	X		X	19
CHANNEL SELEC-TION TIME	3.3.11	4.6.3	X		X	20

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TABLE II. Tests. (continued)

	REQ. PARA	TEST PARA	FIRST ART. TESTS	PRODUC-TION TESTS	SAMPLE TESTS	TEST TEST NO.
RESETTABILITY	3.3.10	4.6.3	X		X	21
CHANNEL LOCK-ON TIME	3.3.12	4.6.3	X		X	22
WARMUP TIME	3.3.14	4.6.3	X		X	23
SPURIOUS & HARMONIC RADIATION	3.3.27.5	4.6.2	X		X	24
HIGH AND LOW VOLTAGE OPERATION	3.3.18 3.3.19	4.6.3	X		X	25
ANTENNA PERFORMANCE	3.3.29	4.6.3	X		X	26
CONTINUOUS DUTY	3.3.15	4.6.3	X		X	27
FREQUENCY STABILITY	3.3.27.4	4.6.2	X		X	28
BATTERY LIFE	3.3.32.1	4.6.3	X			29
LOW TEMPERATURE	3.6.3	4.6.4.1	X		X	30
HIGH TEMPERA-TURE	3.6.3	4.6.4.2	X		X	31
ALTITUDE	3.6.3	4.6.4.3	X		X	32
TEMPERATURE SHOCK	3.6.3	4.6.4.4	X		X	33
LEAKAGE	3.6.3	4.6.4.5	X	X	X	34
DUST	3.6.3	4.6.4.6	X		X	35
TRANSIT DROP	3.6.3	4.6.4.7.1	X		X	36
DROP	3.6.3	4.6.4.7.2	X		X	37
BENCH HANDLING	3.6.3	4.6.4.7.3	X		X	38
VIBRATION	3.6.3	4.6.4.8	X		X	39
RAIN (OPERATING)	3.6.3	4.6.4.9	X		X	40
HUMIDITY	3.6.3	4.6.4.10	X		X	41
FUNGUS	3.2.3.3.6.3	4.6.4.11	X		X	42
SALT FOG	3.6.3	4.6.4.12	X		X	43
ELECTROMAGNETIC INTERFERENCE	3.7	4.6.5	X			44
RELIABILITY	3.4	4.6.6	X			45
MAINTAINABILITY	3.5	4.6.7	X			46
INTERCHANGE-ABILITY	3.3.38	4.6.8	X		X	47

4.4 Production inspection. All radio sets shall be subjected to the tests indicated for production in Table II, in accordance with Drawing No. 671-3412.

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4.4.1 Preproduction inspection. Prior to production inspection, each radio set shall be subjected to the temperature cycling and workmanship screen specified below:

4.4.1.1 Temperature cycling. Each radio set shall be subjected to the temperature curve indicated in Drawing No. 671-3412. The number of cycles required shall be determined by the electric/electronic parts count and the parts count chart in Table III.

TABLE III. Electric/electronic parts count.

ELECTRIC/ELECTRONIC PARTS COUNT.	NO. OF TEMPERATURE CYCLES
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

4.4.1.1.1 Failures during temperature cycling. Failures occurring prior to the last original temperature cycle shall be repaired and the cycling continued. If failures occur during the last original temperature cycle, Table IV shall be used to determine the number of additional cycles required.

TABLE IV. Equipment parts count.

Percentage of Total Parts Reworked/Replaced 1/	Equipment Parts Count		
	500 or less	501 to 2000	2000 and up
up to 0.1 percent	None	repeat 1 cycles	repeat 1 cycle
0.1 to 1 percent	repeat 1 cycle	repeat 1 cycles	repeat 2 cycles
1 to 5 percent	repeat 2 cycles	repeat 2 cycles	repeat 4 cycles
5 to 10 percent	repeat 2 cycles	repeat 4 cycles	repeat 6 cycles
	Last cycle shall be failure free		

1/ Each reworked solder joint or resoldered lead shall count as one replaced.

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4.4.1.2 Workmanship screen. Prior to conducting temperature cycling test, vibration shall be performed on each radio set. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration. Refer to Drawing No. 671-3412 for the procedures to be performed.

4.5 Sample inspection. Tests indicated for sample in Table II shall be performed upon samples from production lots as determined by the following sampling:

LOT SIZE	QUANTITY TO BE TESTED	SAMPLE NO.
25	3	Lot 1
75	3	Lot 2
75 or portion thereof	3	Lot 3
75 or portion thereof	3	Lot 4

Sample environmental tests from Table II shall be divided into three sets as given by the following:

SET No.	TEST No.
1	30, 31, 32, 33, 41
2	34, 36, 37, 38, 39
3	35, 40, 43, 47

Each radio set from the three samples selected for environmental testing shall be subjected to only one set of tests. Radio sets or components thereof shall have successfully passed the individual production tests prior to selection for sample testing. Failure of the sample shall constitute rejection of the lot. Failure is defined as the inability of the radio set to meet the requirements specified herein. The details of the tests to be performed shall be called out in the sample test procedures.

4.5.1 Noncompliance. If a sample unit fails the tests specified in the sample test column of Table II, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of production which were manufactured with the same conditions, materials and processes. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. Approval to ship the lot may be withheld at the discretion of the Government, pending the

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decision from the contracting officer on the adequacy of corrective action. The Government reserves the right to require selection of new samples and a new sample test be performed per Table II after completion of any reworking of the lot.

4.5.2 Disposition of tested equipment. Upon successful completion of the sample tests, the test samples shall be returned to the production area, refurbished as required and resubjected to the production tests of Table II. Equipments which are subjected to, and pass, this production inspection and meet the requirements specified herein may be deliverable as part of the contract quantity.

4.6 Inspection and tests.

4.6.1 Mechanical inspection. The radio set shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements specified herein and MIL-STD-454. Particular attention shall be given to the following:

a. Completeness.
b. Nameplates, identification markings and labels.
c. Ease of operation of gears, adjustable and sliding parts, thumb screws, controls and switches.
d. Finish.
e. Welded joints.
f. Check of solder joints.
g. The fit of components in their respective positions.
h. Check of mounting means.
i. Check of lubrication and rust prevention.
j. Check of safety features and interlocks.
k. Loose fastening and securing devices or parts.
l. Accessibility of components and parts for servicing.
m. Cable run between components including plugs and receptables.
n. Overall dimensions check.
o. Weight check.
p. Any visible defects.

4.6.2 Electrical inspection. The radio set shall be given all electrical tests necessary to confirm that all circuits are inherently sound and in compliance with this specification. The electrical tests shall include, but not necessarily be restricted to, the following:

a. Continuity. Each electrical component and cable shall be given continuity tests as necessary to ascertain that they are wired and connected correctly and that good electrical contact is obtained.

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b. Operating voltages. The operating voltage at all important points shall be checked for conformance to those shown on the circuit label and schematic drawing. This shall be done with all controls set for normal operation of the radio set.

c. Electrical measurements. Any measurements from module pins, connector terminals and other pertinent points to ground shall be made with all controls set for normal operation of the radio set. Any resistance measurements from the above points to ground shall be made with all controls set for normal operation of the radio set, but with no power applied. Instruments of high input impedance shall be used for these tests.

4.6.3 Performance tests. First article and sample radio sets shall be given the performance tests indicated in Table II to confirm that the radio sets meet the performance specified herein.

4.6.4 Environmental tests. The radio sets shall be subjected to the following environmental tests performed in accordance with MIL-STD-810 in the sequence specified by Table I, Group 1E of MIL-STD-810. Performance tests 5 through 10 listed in Table II shall be performed and used as the data base to determine compliance with the requirements of 3.6.

4.6.4.1 Low temperature. The low temperature test shall be conducted in accordance with Method 502, procedure I, of MIL-STD-810. The storage temperature shall be -54°C. The operating temperature shall be -30°C and an external power source may be used.

4.6.4.2 High temperature. The high temperature test shall be in accordance with Method 501, procedure I, of MIL-STD-810. The radio set, less battery, shall be tested for storage at a temperature of +68°C and for operation at a temperature of +52°C. The battery will not be required to pass the environmental requirements of this specification.

4.6.4.3 Altitude. The altitude test shall be conducted in accordance with method 500, procedure I, of MIL-STD-810 on the radio set, less battery.

4.6.4.4 Temperature shock. The temperature shock test shall be conducted in accordance with Method 503, procedure I, of MIL-STD-810, except the temperature shall be +68°C for steps 1 and 4.

4.6.4.5 Leakage (immersion). The leakage test shall be conducted in accordance with Method 512, procedure I, of MIL-STD-810.

4.6.4.6 Dust. The radio set, in operating condition, shall be subjected to dust test in accordance with Method 510, procedure I, of MIL-STD-810.

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4.6.4.7 Shock. The shockproofness of the radio set shall be determined by the following tests.

4.6.4.7.1. Transit drop. The transit drop test shall be conducted in accordance with method 516, procedure II, of MIL-STD-810.

4.6.4.7.2 Drop. The RT and battery shall each be capable of withstanding a drop to a flat surface of solid wood, 1 inch thick fir, backed by at least 3 inches of concrete, along each of three mutually perpendicular planes from a height of 42 inches. The battery (see 3.3.32) shall be within the battery case when the radio set is subjected to the drop test. After the drop test, the RT shall perform as specified herein; the RT may be repaired as specified in 3.5.1 if necessary.

4.6.4.7.3 Bench handling. The RT shall be tested for bench handling in accordance with method 516, procedure V, of MIL-STD-810.

4.6.4.8 Vibration. The RT shall be subjected to a vibration test in accordance with method 514.2, procedure VIII, CURVE Y of figure 514.2-6 and part 2 of procedure XI of MIL-STD-810.

4.6.4.9 Rain (operating). The rain test shall be in accordance with method 506, procedure I, of MIL-STD-810.

4.6.4.10 Humidity. The humidity test shall be in accordance with method 507, procedure I, of MIL-STD-810, except the high temperature shall be +68°C.

4.6.4.11 Fungus. The radio set shall be tested for fungus-proofness in accordance with method 508, procedure I, of MIL-STD-810.

4.6.4.12 Salt fog. The salt fog test shall be conducted in accordance with method 509, procedure I, of MIL-STD-810 and shall be performed with the radio set in operating condition. Salt deposits may be removed by rising with tap water prior to operation. Exposure shall be continued for 100 hours.

4.6.5 EMI tests. The RT shall be tested for EMI in accordance with MIL-STD-461 for class IA and IB equipment, with the following exceptions:

a. Transmitter fundamental and harmonics of the fundamental shall be exempted from the requirements placed upon radiated interference from case and cabling (Method RE02 of MIL-STD-462).

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b. During receiver operation, and without use of the antenna adaptor, antenna-conducted continuous wave (CW) interference signals shall not exceed 50 dB above 1 microvolt. With use of the antenna adaptor, antenna conducted CW interference signals above 450 MHz band shall not exceed 40 dB above 1 microvolt.

c. Rejection of undesired signals. The limit for rejection of undesired signals within the tuning band of the RT shall be 60 dB in lieu of 80 dB as called out for limit A in Figure 18 of MIL-STD-461. The limit for undesired signals outside the tuning band of the RT shall be -30 dB mW in lieu of 0 dB mW as called out for limit B in Figure 18 of MIL-STD-461.

d. Cross modulation. The level of the undesired signal above the desired signal reference at which the signal-to-noise ratio degrades from 10 dB to 6 dB shall be not less than 50 dB. Undesired signals within the specified 60 dB bandwidth of the receiver shall be excluded from the test.

e. Spike and squelch susceptibility. The RT shall be exempted from the spike and squelch susceptibility requirements of this specification. Squelch susceptibility shall be tested and reported for information purposes only.

f. Radiated interference during receive operation. Narrow band radiation interference signals above 100 MHz shall not exceed the levels specified in Figure 21 of MIL-STD-461 by more than 15 dB.

g. Conducted interference and susceptibility on power and signal leads. The radio set (RT and battery) shall be tested in accordance with the methods of MIL-STD-462. Test results shall be reported for possible considerations necessary for interface requirements in vehicular and aircraft installations.

h. Intermodulation. The level for intermodulation responses of the RT shall be 50 dB in lieu of 66 dB as called out for limit CS03 of MIL-STD-461.

4.6.6 Reliability test. Reliability tests to demonstrate first article conformance to 3.5 shall be performed in accordance with MIL-STD-781, test plan VII C, except that the temperature range shall be -30°C to +52°C. Temperature cycling, voltage cycling, and duty cycling shall conform to MIL-STD-781. For purposes of reliability testing, the duty cycle shall be 9 minutes receive and 1 minute transmit for each 10 minutes of operation. The first articles shall be subjected to the reliability test program as specified herein, immediately after undergoing the mechanical, electrical, performance,

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and environmental tests required in 4.6.1, 4.6.2, 4.6.3, and 4.6.4. Electrical performance of the radio set shall not be affected by any test applied. Failures shall be analyzed in accordance with MIL-STD-781.

4.6.7 Maintainability test. The maintainability test to demonstrate the MTTR specified in 3.5.1 shall be in accordance with MIL-STD-470, Method 2.

4.6.8 Interchangeability test. All modules of the RT shall be tested for interchangeability by substitution of like modules between main chassis and control head modules (1A1). An operational checkout by establishing communications between two radio sets on at least three frequencies will confirm the interchangeability of modules.

5. PACKAGING

5.1 Preservation, packaging and packing. Materiel being prepared for shipment shall be afforded the protection required to prevent deterioration or damage due to hazards known, or anticipated during periods of shipment and storage.

5.2 Level of protection. Appropriate military levels of protection or commercial packaging shall be specified. Military levels shall be in accordance with MIL-E-17555 and commercial packaging in accordance with ASTM D3951-82.

5.3 Marking. In addition to any special markings required by the contract or order, unit packages, intermediate packages, and shipping containers stipulating military packaging shall be marked in accordance with MIL-STD-129. Commercial packaging shall be marked in accordance with ASTM D3951-82.

5.4 Special requirements. If any of the subject items or components associated with them are identified as Electrostatic Discharge Sensitive Devices they shall be packaged in accordance with the provisions of DOD-HDBK-263 and DOD-STD-1686.

6. NOTES

6.1 Intended use. The radio set covered by this specification is a small, lightweight, AM UHF RT which will be subject to extremely rough field usage by infantry personnel.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number and date of this specification.

b. When first article is not required (see 3.1).

c. Levels of preservation, packaging and packing (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a Contract Data Requirements List (CDRL) (DD Form 1423), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract purchase order requirements. Deliverable data required by this Specification is cited in the following paragraphs:

Para No.	Data Requirement Title.	Applicable DD No Option.
3.7	Electromagnetic Interference Control Plan	DI-E-1128
4.6.6	Electromagnetic Interference Test Plan	DI-R-7063
4.3	First Article Inspection Procedures	DI-T-4901
4.3	First Article Inspection Report	DI-T-4902
4.5	Environmental Program - Pretest Performance Record	DI-T-4906

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such as DOD 5000.19L, Volume II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.

6.3 First article instructions. The first articles shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory, acceptable to the procuring activity. Selected first article tests will be witnessed by a representative of the acquiring activity. Test witnessed by a representative of the procuring activity shall be witnessed and certified by the appropriate Defense Contract Administration Service District representative. Approval of the first articles shall be by the contracting officer.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.12.

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6.5 Subject term (key word) listing.

Antenna
Battery Box
Battery, Dry
Cable Assembly
Case, Radio Set
Handset
Harness, Radio Set
Headset
Modular Concept
Radio Set
Receiver - Transmitter
Solid State components

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue date due to the extensiveness of the changes.

Preparing Activity:
Navy - MC
Project No. 5820-N797

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approximately 70 KVP shall be given. Sur-

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-82196B(MC)	RADIO SET: AN/PRC-75B

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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